

Question description

There are n boys and m girls in a school. Next week a school dance will be organized. A dancing pair consists of a boy and a girl, and there are k potential pairs.

Your task is to find out the maximum number of dance pairs and show how this number can be achieved.

Constraints

- $1 \leq n, m \leq 500$
- $1 \leq k \leq 1000$
- $1 \leq a \leq n$
- $1 \leq b \leq m$

Input

The first input line has three integers n , m and k : the number of boys, girls, and potential pairs. The boys are numbered $1, 2, \dots, n$, and the girls are numbered $1, 2, \dots, m$.

After this, there are k lines describing the potential pairs. Each line has two integers a and b : boy a and girl b are willing to dance together.

Output

First print one integer rr : the maximum number of dance pairs. After this, print lines describing the pairs. You can print any valid solution.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

3 2 4
1 1
1 2
2 1
3 1

EXPECTED OUTPUT

2
1 2
2 1

Test Case 2

INPUT (STDIN)

3 2 4
1 1
1 2
2 3
3 2

EXPECTED OUTPUT

3
1 1
2 3
3 2

▼ Mandatory Test Cases

Test Case 1

KEYWORD

void link(int u,int v)

Test Case 2

KEYWORD

int bfs(int n)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

600

Test Case 3

NLOC

84